

8.2.37

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September 18, 2025

Question

Find the equation of the conic, that satisfies the given conditions. vertex $(-3, 0)$, directrix $x + 5 = 0$.

Theoretical Solution

Given:

- Vertex: $\mathbf{V} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}$
- Directrix: $x + 5 = 0$, which gives us $\mathbf{n}^\top \mathbf{x} = c$ where $\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $c = -5$

The general matrix equation of a conic is:

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1)$$

where the matrices are defined as:

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2(\mathbf{n}\mathbf{n}^\top) \quad (2)$$

$$\mathbf{u} = (ce^2)\mathbf{n} - \|\mathbf{n}\|^2 \mathbf{F} \quad (3)$$

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2 e^2 \quad (4)$$

Theoretical Solution

For a vertex at $(-3, 0)$ and using the vertex-focus-directrix geometry, the focus $\mathbf{F}$ is at:

$$\mathbf{F} = \begin{pmatrix} -3 + 2e \\ 0 \end{pmatrix} \quad (5)$$

Case 1: $e < 1$ (Ellipse)

Let $e = \frac{1}{2}$

Parameters:

- $\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $c = -5$, $e = \frac{1}{2}$
- $\mathbf{F} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$
- $\|\mathbf{n}\|^2 = 1$, $\|\mathbf{F}\|^2 = 4$

Theoretical Solution

Matrix Calculation:

$$\mathbf{v} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{4} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 3/4 & 0 \\ 0 & 1 \end{pmatrix} \quad (6)$$

$$\mathbf{u} = \frac{-5}{4} \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} -2 \\ 0 \end{pmatrix} = \begin{pmatrix} 3/4 \\ 0 \end{pmatrix} \quad (7)$$

$$f = 4 - \frac{25}{4} = -\frac{9}{4} \quad (8)$$

Putting Values of $\mathbf{V}$, $\mathbf{u}$, f in Equation 0.1, we get

$$\frac{3}{4}x^2 + y^2 + \frac{3}{2}x - \frac{9}{4} = 0 \quad (9)$$

$$\boxed{\frac{(x+1)^2}{4} + \frac{y^2}{3} = 1} \quad (10)$$

Theoretical Solution

Case 2: $e = 1$ (Parabola)

Parameters:

- $\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $c = -5$, $e = 1$
- $\mathbf{F} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$
- $\|\mathbf{F}\|^2 = 1$

Matrix Calculation:

$$\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (11)$$

$$\mathbf{u} = -5 \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \end{pmatrix} \quad (12)$$

$$f = 1 - 25 = -24 \quad (13)$$

Theoretical Solution

Putting Values of $\mathbf{V}$, $\mathbf{u}$, f in Equation 1, we get

$$y^2 - 8x - 24 = 0 \quad (14)$$

OR

$$\boxed{y^2 = 8(x + 3)} \quad (15)$$

Theoretical Solution

Case 3: $e > 1$ (Hyperbola)

Let $e = \frac{3}{2}$

Parameters:

- $\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $c = -5$, $e = \frac{3}{2}$
- $\mathbf{F} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$
- $\|\mathbf{F}\|^2 = 0$

Matrix Calculation:

$$\mathbf{v} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{9}{4} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} -5/4 & 0 \\ 0 & 1 \end{pmatrix} \quad (16)$$

$$\mathbf{u} = \frac{-45}{4} \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -45/4 \\ 0 \end{pmatrix} \quad (17)$$

$$f = 0 - \frac{225}{4} = -\frac{225}{4} \quad (18)$$

Putting Values of **V**, **u**, **f** in Equation 1, we get

$$-\frac{5}{4}x^2 + y^2 - \frac{45}{2}x - \frac{225}{4} = 0 \quad (19)$$

$$\boxed{\frac{(x + 9)^2}{36} - \frac{y^2}{45} = 1} \quad (20)$$

Theoretical Solution

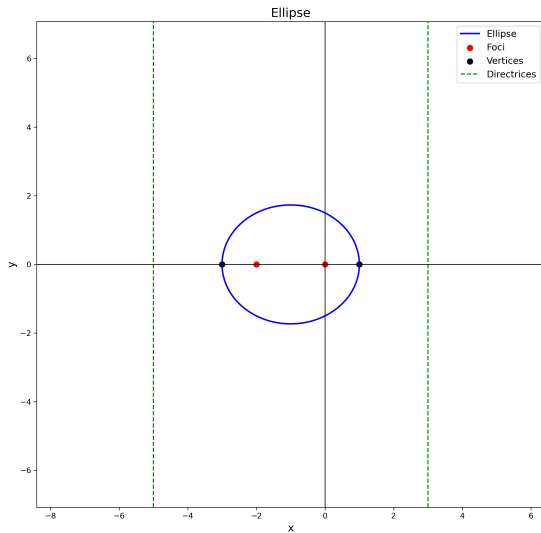
Therefore, The Possible conics with the vertex at $(-3,0)$ and Directrix as $x+5=0$ are

- $\frac{(x+9)^2}{36} - \frac{y^2}{45} = 1$

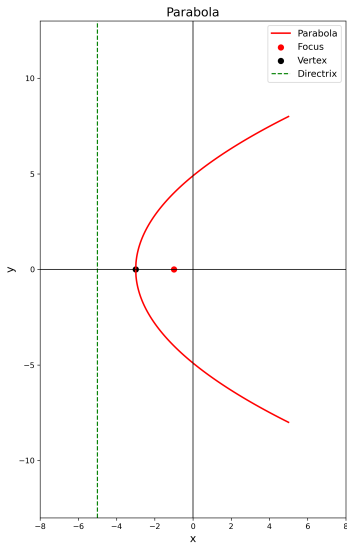
- $y^2 = 8(x+3)$

- $\frac{(x+1)^2}{4} + \frac{y^2}{3} = 1$

Ellipse



Parabola



Hyperbola

